SOC DETECTION LAB

Master Architecture & Operations Document

Controlled environment with VLAN-segmented network

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1. Executive Summary

This document describes the architecture, configuration, resolved incidents, and operational status of a home Security Operations Center (SOC Lab) built in a virtualized environment using VMware on a physical iMac host.

The lab simulates a real enterprise infrastructure, including VLAN network segmentation, dual SIEM platforms, Active Directory, endpoint telemetry with Sysmon, network intrusion detection with Suricata, and attack simulation systems.

Lab Objectives

- Practice SOC analyst workflows in a controlled, realistic environment
- Operate and administer dual SIEM platforms (Wazuh and Splunk) simultaneously
- Simulate internal and external attacks to validate detection capabilities
- Investigate alerts and correlate events from multiple telemetry sources
- Develop skills in forensic analysis, threat hunting, and incident response

Current Lab Status

- **Fully segmented VLAN architecture — operational**
- Wazuh SIEM active with agents deployed on all endpoints
- Splunk SIEM active receiving telemetry from Windows, Linux, and physical host
- Sysmon deployed on all Windows endpoints
- Suricata IDS running on pfSense for network traffic analysis
- Active Directory with LAB.local domain and test user accounts

- Ubuntu Data Server joined to domain via Kerberos/realm
- Physical iMac host forwarding logs to both SIEMs through pfSense

2. Network Architecture

The lab is divided into two primary segments: the home WAN network where the physical host resides, and the internal lab network composed of multiple VLANs segmented through pfSense.

2.1 WAN — Physical Host

Network	192.168.0.0/24 — TP-Link Router (home network)
Physical Host	iMac — macOS — Runs VMware hosting all VMs
Function	The iMac forwards logs to Wazuh and Splunk through pfSense
Future Phase	Red Team Kali machines will connect here in bridge mode to simulate real external attacks

2.2 LAN — VLAN Segmentation

All inter-VLAN traffic passes through pfSense, enabling complete inspection, security rule enforcement, and Suricata analysis across all segments.

VLAN	Name	Subnet	Systems	VMnet
10	Management	10.10.10.0/24	pfSense, Suricata, WIN11-WS	VMnet3
20	Users	10.10.20.0/24	USER01, USER02	VMnet4
30	Executive	10.10.30.0/24	USER03, USER04	VMnet5
40	Servers	10.10.40.0/24	DC01, Data Central	VMnet6
50	SOC	10.10.50.0/24	Wazuh, Splunk, Kali Blue	VMnet7
60	Quarantine	10.10.60.0/24	N/A (Reserved)	VMnet9

2.3 pfSense Firewall

WAN Interface	192.168.0.x — connected to the home TP-Link router network
LAN Interface	10.10.10.1 — gateway for VLAN 10 (Management)

Functions	Inter-VLAN routing, NAT, firewall rules, Suricata IDS host

pfSense acts as the network core of the lab. All VMs communicate through pfSense, enabling full traffic control and monitoring. Suricata runs directly on pfSense analyzing inter-VLAN traffic.

3. Security Stack

The lab implements multiple security layers that replicate a real enterprise environment. Each tool serves a specific role within the detection and response pipeline.

Tool	Category	Primary Function
pfSense	Firewall / Router	VLAN segmentation, inter-VLAN routing, traffic control
Suricata	Network IDS	Malicious traffic detection, network anomaly analysis
Wazuh	SIEM / XDR	Log centralization, event correlation, threat detection, alerts
Splunk	SIEM / Analytics	Log indexing, SPL search, dashboards, correlation
Sysmon	Endpoint Telemetry	Advanced events: processes, network, registry, files
Active Directory	Identity Management	Centralized authentication, GPOs, LAB.local domain
VirusTotal API	Threat Intelligence	Automatic hash reputation enrichment for suspicious files
Kali Blue	Blue Team	Analysis, SSH access to CLI servers, detection validation

3.1 Wazuh — SIEM / XDR

IP	10.10.50.20 — VLAN 50 (SOC)

System	Ubuntu Server (no GUI)

Ports	1514 (log reception) / 1515 (agent enrollment)

Wazuh serves as the central monitoring platform. It receives telemetry from all endpoints via installed agents. Log source configuration is managed in /var/ossec/etc/ossec.conf.

Log sources configured in Wazuh:

- Windows Security Event Logs

- Microsoft-Windows-Sysmon/Operational
- Physical host iMac logs
- pfSense firewall logs

VirusTotal API integration: when Wazuh detects suspicious files, it automatically queries the hash reputation (MD5, SHA1, SHA256) against VirusTotal antivirus engines.

3.2 Splunk — SIEM / Analytics

IP	10.10.50.10 — VLAN 50 (SOC)

System	Ubuntu Server (no GUI, accessed via Kali Blue SSH)

Ports	8000 (Web UI) / 9997 (Forwarder ingestion)

Splunk complements Wazuh with advanced search capabilities using SPL (Search Processing Language), event correlation, and dashboards. Splunk Universal Forwarders are deployed on all endpoints.

Endpoints integrated in Splunk:

- ADMIN-WORKSTATION (Windows 11)
- SERVER-LAB / DC01 (Windows Server 2025)
- WIN11-USER01 and WIN11-USER02
- EXEC-USER03 and EXEC-USER04
- Kali Blue (jean-blue) — Linux
- iMac physical host — macOS

3.3 Suricata — Network IDS

Location	Running on pfSense — VLAN 10 (Management)

Suricata analyzes all traffic passing through pfSense between VLANs. Being integrated directly into the firewall, it has full visibility of inter-VLAN communications, enabling detection of network scans, malicious traffic, and anomalies.

3.4 Sysmon — Endpoint Telemetry

Sysmon is deployed on all Windows endpoints in the lab. A custom configuration file (sysmonconfig.xml) based on community rules is recommended to capture the most security-relevant events.

Key events captured by Sysmon:

- EventID 1 — Process Creation
- EventID 3 — Network Connection
- EventID 11 — File Creation
- EventID 13 — Registry Value Set
- EventID 7 — Image Loaded

4. Infrastructure Details

4.1 Active Directory — DC01

IP	10.10.40.10 — VLAN 40 (Servers)
System	Windows Server 2025
Domain	LAB.local
Functions	Centralized authentication, GPOs, domain DNS, access control

Test groups and user accounts:

- Group LAB-Users: USER01, USER02 (VLAN 20)
- Group EXEC-Users: USER03, USER04 (VLAN 30)

4.2 Ubuntu Data Central Server

IP	10.10.40.x — VLAN 40 (Servers)
System	Ubuntu Server (no GUI, CLI access only)
AD Integration	Joined to LAB.local domain via realm + Kerberos

The Ubuntu server was integrated into Active Directory to enable centralized authentication across a hybrid Windows/Linux infrastructure. Integration required NTP time synchronization for Kerberos and DNS configuration pointing to DC01.

4.3 User Endpoints

All Windows endpoints have installed: Wazuh Agent, Splunk Universal Forwarder, and Sysmon.

USER01	Windows 11 — VLAN 20 — Generates normal domain activity

USER02	Windows 11 — VLAN 20 — Generates normal domain activity
USER03 (EXEC)	Windows 11 — VLAN 30 — Simulates executive user
USER04 (EXEC)	Windows 11 — VLAN 30 — Simulates executive user

4.4 Admin Workstation

IP	10.10.10.x — VLAN 10 (Management)
System	Windows 11
Function	Domain administration, GPO management, general monitoring

4.5 Kali Blue Team

Hostname	jean-blue — VLAN 50 (SOC)
Function	Blue team analysis, SSH access to CLI servers (Splunk, Data Central), detection validation

4.6 Kali Red Team (Future Phase)

Planned Location	WAN network 192.168.0.0/24 — VMware bridge mode

Red Team Kali machines will be activated once the lab is fully configured. They simulate real external attackers connected directly to the same network as the pfSense WAN interface, performing reconnaissance, scanning, and exploitation from outside the network perimeter.

5. Telemetry & Log Pipeline

Log flow from endpoints to the SIEMs follows a layered model at each stage of the pipeline.

5.1 Flow from Windows Endpoints

1. User activity generates events in the Windows Event Log
2. Sysmon captures advanced telemetry events (processes, network, files)
3. Wazuh Agent collects and forwards events to Wazuh Manager (10.10.50.20:1514)
4. Splunk Universal Forwarder sends logs to Splunk Indexer (10.10.50.10:9997)

5. Both SIEMs process, index, and generate alerts

5.2 Flow from Linux / Kali Blue

6. rsyslog generates log files: /var/log/syslog, /var/log/auth.log, /var/log/kern.log
7. Splunk Universal Forwarder monitors those files and forwards to the indexer
8. Wazuh Agent collects Linux system logs

5.3 Flow from Physical Host (iMac)

9. iMac generates macOS system logs
10. Forwarder on the iMac sends logs through pfSense
11. pfSense allows traffic to VLAN 50 (port 1514 for Wazuh, 9997 for Splunk)
12. Wazuh Manager and Splunk Indexer receive the events

5.4 Log Sources by SIEM

Source	Wazuh	Splunk
Windows Security Logs	Yes	Yes
Sysmon (XML Event Log)	Yes	Yes
Windows System/App Logs	Yes	Yes
Linux syslog / auth.log	Yes	Yes
pfSense Firewall Logs	Yes	Partial
macOS Host Logs (iMac)	Yes	Yes
Suricata IDS Events	Via pfSense	Pending

6. Resolved Incidents

This section documents technical issues encountered during the construction and operation of the lab. Each incident includes a description, root cause, applied solution, and lesson learned.

Incident 1 — Access Denied on Sysmon Channel (Splunk)

Field	Value
Date	March 2026 — Version 3.0
Environment	VMware Lab — VLAN 50
Affected Systems	EXEC-USER03, EXEC-USER04
Severity	Medium — No Sysmon telemetry in SIEM
Status	RESOLVED

Description

EXEC-USER03 and EXEC-USER04 were not sending Sysmon events to Splunk. Sysmon was installed and generating events locally, the SplunkForwarder service was Running, connectivity to port 9997 was successful, and inputs.conf was correctly configured.

Root Cause

The Splunk Universal Forwarder runs under the virtual account NT SERVICE\SplunkForwarder. This account does not have permission to subscribe to the Microsoft-Windows-Sysmon/Operational channel. Sysmon creates this channel with stricter ACLs than standard Windows event channels.

Error identified in splunkd.log:

WinEventLogChannel::init: Init failed — errorCode=5 (Access Denied)

Solution Applied

13. Add the service account to the Event Log Readers group:

Add-LocalGroupMember -Group "Event Log Readers" -Member "NT SERVICE\SplunkForwarder"

14. Restart the forwarder service:

Restart-Service SplunkForwarder

15. Verify in Splunk with SPL:

index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" | stats count by host

Prevention

- Include Add-LocalGroupMember in the installation script for every new Windows VM
- Create a GPO in Active Directory to automatically apply this permission to all domain machines
- Add a Splunk alert to detect hosts that stop sending events for more than 15 minutes

Lesson Learned

Windows event channels have independent ACLs. Security telemetry channels (Sysmon) have additional restrictions that do not apply to standard channels. The pipeline must be validated step by step: local generation, forwarder permissions, connectivity, SIEM ingestion.

Incident 2 — tcpout Queue Blocked by VMware Time Jump

Field	Value
Environment	VMware Lab
Affected Systems	EXEC-USER03, EXEC-USER04
Severity	Low — Ingestion delay
Status	RESOLVED

Description

When resuming suspended VMs in VMware, the system clock jumped forward approximately 17 minutes. This generated an invalid -1 value (represented as 18446744073709551615 as an unsigned integer) in the Splunk tcpout queue, blocking log transmission.

Solution Applied

Configure VMware Tools to synchronize the clock when resuming each VM. Add the following to each VM's .vmx file:

tools.syncTime = "TRUE"

Lesson Learned

In virtualized environments, time synchronization is critical for the correct operation of security tools. Clock jumps can cause failures in ingestion pipelines and compromise timestamp integrity across logs.

Incident 3 — Active Directory Trust Relationship Failure

Description

Endpoints displayed the error: 'The trust relationship between this workstation and the primary domain failed.' Machines could not authenticate against the LAB.local domain.

Root Cause

Endpoints had the pfSense gateway (10.10.X.1) configured as their DNS server instead of the Domain Controller (10.10.40.10). Without correct DNS resolution to the DC, Kerberos authentication and the domain secure channel both failed.

Solution Applied

16. Update DNS on endpoints to point to DC: 10.10.40.10
17. Flush DNS cache: ipconfig /flushdns
18. Repair the secure channel from PowerShell: Test-ComputerSecureChannel -Repair
19. Validate with: gpupdate /force

Lesson Learned

In a VLAN architecture, endpoint DNS must always point to the Domain Controller, not the gateway. This is a common mistake when migrating from a flat network to a segmented architecture.

Incident 4 — Wazuh Agents Disconnected After IP Change

Description

After migrating Wazuh Manager to the new VLAN architecture (new IP: 10.10.50.20), all Windows agents appeared as DISCONNECTED in the Wazuh dashboard.

Root Cause

The ossec.conf file on each endpoint still had the old Wazuh Manager IP configured. Agents were attempting to connect to an address that no longer corresponded to the server.

Solution Applied

20. Locate and edit ossec.conf on each endpoint
21. Update the manager IP to 10.10.50.20
22. Restart the Wazuh Agent service
23. Verify ACTIVE status in the dashboard
24. Generate a test event to confirm telemetry reception

Lesson Learned

When migrating SIEM infrastructure to a new network or IP, all agents must be updated. In production, this would be handled via a mass update script or Group Policy.

Incident 5 — Splunk Forwarders Pointing to Old IP Post-VLAN Migration

Description

After the VLAN migration, Splunk Universal Forwarders were still pointing to the old Splunk server IP (10.10.0.30). Endpoints were not appearing in the Splunk host list.

Solution Applied

25. Update outputs.conf on each endpoint: server = 10.10.50.10:9997
26. Restart the SplunkForwarder service
27. Verify connectivity: Test-NetConnection 10.10.50.10 -Port 9997
28. Confirm host visibility in Splunk Search & Reporting

Incident 6 — inputs.conf Saved with .txt Extension

Description

The Splunk Forwarder inputs configuration file was accidentally saved as inputs.conf.txt because Windows Explorer hides file extensions by default.

Solution Applied

Rename the file using PowerShell or terminal to ensure the correct extension. Windows Explorer does not allow renaming when extensions are hidden.

Incident 7 — Kali Missing syslog Files (journald vs syslog)

Description

The Kali Blue Team system did not have /var/log/syslog, /var/log/auth.log, or /var/log/kern.log files. Splunk Universal Forwarder reads log files, not the systemd journal.

Solution Applied

29. Install rsyslog: sudo apt install rsyslog
30. Enable the service: sudo systemctl enable --now rsyslog
31. Verify creation of standard log files
32. Configure forwarder inputs.conf to monitor those files

Incident 8 — Ubuntu Domain Join Failure (Time Sync / Kerberos)

Description

When attempting to join the Ubuntu Data Server to the LAB.local domain, Kerberos authentication failed due to a time difference between the Ubuntu server and the Domain Controller.

Solution Applied

33. Synchronize NTP with the Domain Controller
34. Install Kerberos client and realm tools
35. Join the domain: realm join LAB.local
36. Validate authentication: kinit <user>@LAB.LOCAL
37. Verify ticket: klist
38. Confirm AD identity resolution on Linux: id <user>

7. Detection Scenarios Practiced

The following scenarios have been executed and validated in the lab to confirm the SOC detection capabilities.

7.1 Hostname Change Investigation (Forensic Example)

The Kali Blue host was renamed from 'jeanread' to 'jean-blue'. This change was forensically reconstructed using Splunk.

SPL Query used:

```
index=* (host=jeanread OR host=jean-blue)
| search hostnamectl OR hostnamed OR hostname
| table _time host source _raw | sort _time
```

Result: Splunk preserved historical events under the original hostname (jeanread) and new events under jean-blue, enabling complete timeline reconstruction of the administrative change.

7.2 Privilege Escalation Detection (Linux)

Sudo usage detected in Kali Blue Team authentication logs.

SPL Query:

```
index=* sourcetype=linux_secure | search sudo | table _time host _raw | sort -_time
```

7.3 Sysmon Telemetry Validation Post-Fix

After resolving the permissions incident, test events were generated and full Sysmon telemetry ingestion was validated in Splunk.

Test actions executed:

- Execute whoami from PowerShell (EventID 1)
- Create a temporary file (EventID 11)
- Test network connection (EventID 3)

7.4 Local User Creation Detection (Wazuh)

A local user creation event was generated on a Windows endpoint. Wazuh generated the corresponding alert, validating the complete detection pipeline from endpoint to SIEM.

7.5 Planned Scenarios — Red Team Phase

- Network scanning with Nmap from external Kali — detection via Suricata/pfSense
- Suspicious process execution on Windows — detection via Sysmon EventID 1
- Lateral movement with credentials — detection via AD logs (EventID 4648, 4624)
- Domain reconnaissance (net user /domain, whoami /all) — Sysmon + Wazuh

8. Roadmap — Upcoming Improvements

The lab will continue to evolve. The following improvements are planned to bring the SOC Lab to a full professional demonstration standard.

8.1 High Priority Items

Improvement	Description	Status
Kali Red Team (external)	Activate in bridge mode to simulate real external attacks	Planned
SOC Dashboard in Splunk	Panels for Sysmon, firewall, auth events, IDS	Pending
Velociraptor	Threat hunting and DFIR on endpoints	Pending
Autopsy	Digital forensic analysis of disks and artifacts	Pending
Visual Architecture Diagram	Network diagram in draw.io or Lucidchart	Pending
Suricata → Splunk Integration	EVE JSON logs to indexer via forwarder	Pending
3 Documented Attack Scenarios	Full end-to-end attack with detection and report	In Progress
GPO for Event Log Readers	Automatically apply Splunk permissions via AD policy	Pending

8.2 Long-Term Vision

- Implement SOAR platform for automated incident response
- Integrate TheHive / Cortex for incident case management
- Deploy a honeypot in VLAN 60 (Quarantine) to capture attacks
- Implement RBAC using AD groups for granular access control
- Document formal incident response playbooks

9. Appendix — Quick Reference

9.1 Lab IPs and Services

System	IP	VLAN	Service / Port
pfSense (LAN)	10.10.10.1	10 - Mgmt	Web UI: 443
Wazuh Manager	10.10.50.20	50 - SOC	1514 / 1515
Splunk Server	10.10.50.10	50 - SOC	8000 / 9997
DC01 (AD)	10.10.40.10	40 - Servers	DNS: 53 / Kerberos: 88
Data Central	10.10.40.x	40 - Servers	SSH: 22
Kali Blue	10.10.50.x	50 - SOC	SSH: 22
USER01 / USER02	10.10.20.x	20 - Users	RDP: 3389
USER03 / USER04	10.10.30.x	30 - Executive	RDP: 3389
Admin WS	10.10.10.x	10 - Mgmt	RDP: 3389
iMac (physical host)	192.168.0.x	WAN	Forwarder: 1514/9997

9.2 Frequently Used Validation Commands

Wazuh

```
# Check agent status
sudo /var/ossec/bin/agent_control -l
```

Splunk

```
# Verify connected forwarders
splunk list forward-server
# Search events from specific host
index=* host=EXEC-USER04 | head 20
```

Windows (PowerShell)

```
# Check Wazuh service
Get-Service wazuh
# Check Splunk Forwarder service
Get-Service splunkforwarder
# Check Sysmon service
Get-Service Sysmon64
# Test connectivity to Splunk
Test-NetConnection 10.10.50.10 -Port 9997
# Repair domain secure channel
Test-ComputerSecureChannel -Repair
```

10. VirusTotal Integration — Real Evidence

This section documents the VirusTotal integration with Wazuh, confirmed and validated with real evidence generated in the lab on March 19, 2026.

10.1 Integration Configuration

The integration is configured in /var/ossec/etc/ossec.conf with the following parameters:

Name	virustotal
Trigger group	syscheck — activates when syscheck detects new or modified files
Alert format	JSON

Active Response	quarantine_file.sh — automatically moves malicious files to quarantine

Trigger rule	Rule 87105 — Level 12 (Critical)

10.2 Detection and Response Pipeline

The complete flow from detection to automatic quarantine:

39. New/modified file appears in syscheck-monitored directory
40. syscheck detects the file in real time and calculates MD5, SHA1, SHA256
41. wazuh-integratord sends the hash to the VirusTotal API
42. VirusTotal responds with results from 67 antivirus engines
43. If malicious — Rule 87105 Level 12 fires
44. quarantine_file.sh automatically moves the file to /home/jean/quarantine/
45. Wazuh generates alert with full compliance mapping

10.3 Real Evidence — VirusTotal Alert

REAL ALERT — Generated on March 19, 2026 at 16:02:07

Field	Value
Timestamp	2026-03-19T16:02:07.600-0400
Rule ID	87105
Level	12 — CRITICAL
Description	VirusTotal: Alert - 62 engines detected this file
File	/home/jean/wazuh-test/eicar-1773950522.txt
MD5	69630e4574ec6798239b091cda43dca0
SHA1	cf8bd9dfddff007f75adf4c2be48005cea317c62
Positive engines	62 out of 67 (92.5% detection rate)
VT scan date	2026-03-19 18:10:05
Agent	jean-SIEM (ID: 000)

10.4 Automatic Response — Quarantine

File was automatically moved to quarantine with NO human intervention

Field	Value
Detection timestamp	2026-03-19T16:02:06.291-0400

Original file	/home/jean/wazuh-test/eicar-1773950522.txt
Quarantined file	/home/jean/quarantine/20260319_160204_eicar-1773950522.txt
Script executed	quarantine_file.sh
Response time	< 2 seconds from detection
Action	File moved and deleted from original directory

10.5 Alert Compliance Mapping

Framework	Control	Description
PCI DSS	10.6.1 / 11.4	Security event monitoring and intrusion detection
GDPR	IV_35.7.d	Data protection impact assessment
MITRE ATT&CK	T1203 — Execution	Exploitation for Client Execution
NIST 800-53	SI.7	System and information integrity

10.6 Syscheck Monitored Directories

Directory	Monitoring type	Function
/home/jean/wazuh-test	Real-time	Test directory — VirusTotal trigger
/home/jean/quarantine	Real-time	Destination for quarantined files
/etc, /usr/bin, /usr/sbin	Periodic (12h)	System integrity monitoring
/bin, /sbin, /boot	Periodic (12h)	Critical binary monitoring

11. Physical Host Maintenance — iMac

This section documents the diagnosis, identified issues, and solutions applied to the physical host running the lab (iMac), as its stability is critical for continuous SOC Lab operation.

11.1 Host Specifications

Component	Specification
Model	iMac20,1 — iMac 2020
Processor	Intel Core i7-10700K — 8 cores / 16 threads
RAM	64 GB — More than sufficient for the complete lab
Internal storage	Internal SSD — macOS and applications
Lab storage	Samsung T7 2TB external SSD — VMs and lab data
Operating system	macOS 26.3.1 (25D2128)
Hypervisor	VMware Fusion

GPU	AMD Radeon RX 5500 XT (AMDRadeonX6000)

11.2 Incident — System Freeze (WindowServer Crash)

Field	Value
Date	March 18, 2026 — 23:39:32
Affected process	WindowServer (macOS graphics server)
Cause	Watchdog timeout — WindowServer unresponsive for 80 seconds
Thermal pressure	ThermalPressureLevelHeavy (2) — maximum level
Status	RESOLVED with temporary measures

Root Cause

Combination of three factors:

- Dust accumulated on fans and heatsinks — never cleaned in 5-6 years of use
- Apple automatic SMC too conservative — fans at 1260 RPM with CPU at 80°C
- High simultaneous VM load generating sustained heat

Thermal Evidence

Sensor	Before	After	Ideal
CPU die (idle)	83.42°C	56-67°C	45-55°C
CPU Core avg	73-82°C	59-66°C	50-60°C
CPU PECI	81°C	65°C	55-65°C
GPU die	67°C	49°C	45-55°C
Fan speed	1260 RPM	1800-2200 RPM	Auto based on load

11.3 Solutions Applied

1. Macs Fan Control

Installed via Homebrew to control internal fans more aggressively than Apple's automatic SMC.

Applied configuration — based on CPU PECI sensor:

Temperature	RPM	State
60°C	1800 RPM	Start of increase
70°C	2200 RPM	Medium load
80°C	2600 RPM	High load
90°C	3067 RPM	Maximum — critical load

Installation:

```
brew install --cask macs-fan-control
```

2. Storage Connectivity Reorganization

Storage setup optimization to maximize speed and stability:

Device	Port	Speed
Apollo Twin (audio)	Thunderbolt 2 — direct	40 Gbps
USB-C Hub (TV + peripherals)	Thunderbolt 1 — direct	40 Gbps
Samsung T7 2TB (VMs)	USB-A 1 — direct iMac	392 MB/s
HDD Time Machine	USB-A 2 — direct iMac	~120 MB/s
HDD Storage	USB-A 3 — direct iMac	~120 MB/s

Note: The Samsung T7 was tested on different USB-A ports. USB-A port 1 (first from the left on the rear) provides the highest speed: 392 MB/s write / 370 MB/s read — sufficient for stable VM operation.

3. Physical Cleaning — Pending

PENDING — Physical cleaning with electric air duster upon Amazon delivery

A rechargeable USB electric air duster was purchased to clean accumulated dust through the RAM access slot (the only removable panel on the iMac without special tools).

Cleaning procedure:

46. Shut down the iMac completely
47. Unplug the power cable
48. Wait 2-3 minutes
49. Open the RAM access panel
50. Blow in short bursts at different angles
51. Close the panel
52. Reconnect and power on
53. Perform SMC reset (unplug for 15 seconds)
54. Verify temperatures with powermetrics

11.4 Host Monitoring Tools

Tool	Function	Installation
Stats	CPU, RAM, GPU, network monitor in menu bar	brew install --cask stats

Macs Fan Control	Manual fan speed control	brew install --cask macs-fan-control
btop	System monitor in terminal	brew install btop
powermetrics	Real temperatures via SMC (native macOS)	sudo powermetrics --samplers smc
Blackmagic Disk Speed Test	Disk speed test	App Store — free

11.5 Long-Term Recommendations

- Get a Thunderbolt Dock (CalDigit TS3 Plus or similar) to resolve connectivity limitations
- Take the iMac to an Apple technician for professional internal cleaning when possible
- Consider internal SSD upgrade for increased capacity
- Maintain periodic cleaning every 6 months with air duster
- Continuously monitor temperature with Stats during long lab sessions

12. Ubuntu Data Server Integration — Wazuh & Splunk

This section documents the complete integration of the Ubuntu Data Central server (10.10.40.100) into both SIEM platforms of the lab, completing full visibility across the hybrid Windows/Linux infrastructure.

12.1 Data Server Specifications

Hostname	data-server
IP	10.10.40.100 — VLAN 40 (Servers)
Operating System	Ubuntu 24.04.4 LTS — Kernel 6.8.0-106-generic x86_64
AD Domain	LAB.local — joined via realm + Kerberos
Access	SSH from Kali Blue (10.10.50.x) — jean@10.10.40.100

12.2 Wazuh Agent Integration

Wazuh Agent installed, enrolled and ACTIVE — March 19, 2026

Installation Process

55. Add Wazuh repository GPG key
56. Configure Wazuh 4.x apt repository

57. Install wazuh-agent via apt
58. Configure Wazuh Manager IP in ossec.conf
59. Enroll agent with agent-auth
60. Start and enable the service

Commands Executed

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --no-default-keyring \
  --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import
```

```
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] \
  https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee \
  /etc/apt/sources.list.d/wazuh.list
```

```
sudo apt update && sudo apt install wazuh-agent -y
```

```
sudo sed -i 's/MANAGER_IP/10.10.50.20/' /var/ossec/etc/ossec.conf
```

```
sudo /var/ossec/bin/agent-auth -m 10.10.50.20
```

```
sudo systemctl enable wazuh-agent && sudo systemctl start wazuh-agent
```

Manager Configuration

Manager IP	10.10.50.20 — VLAN 50 (SOC)
Port	1514 (TCP) — log reception
Enrollment port	1515 — agent authentication
Status	ACTIVE — verified in Wazuh Dashboard
Version	4.14.x — latest available version

12.3 Splunk Universal Forwarder Integration

Splunk Forwarder installed and sending logs to 10.10.50.10:9997

Installation Process

61. Download Splunk Universal Forwarder 9.4.0 for Linux
62. Install via dpkg
63. Configure forwarding to Splunk Indexer

64. Configure log sources in inputs.conf
65. Start and verify the service

Commands Executed

```
wget -O /tmp/splunkforwarder.deb \

https://download.splunk.com/products/universalforwarder/releases/9.4.0/linux/splunkforwarder-9.4.0-6b4ebe426ca6-linux-amd64.deb
```

```
sudo dpkg -i /tmp/splunkforwarder.deb
```

```
sudo /opt/splunkforwarder/bin/splunk add forward-server 10.10.50.10:9997 -auth admin:Admin1234!
```

inputs.conf Configuration

```
[monitor:///var/log/syslog]
disabled = 0
sourcetype = syslog
```

```
[monitor:///var/log/auth.log]
disabled = 0
sourcetype = linux_secure
```

```
[monitor:///var/log/kern.log]
disabled = 0
sourcetype = linux_kernel
```

Validation in Splunk

SPL query to confirm log reception:

```
index=* host=data-server | head 20
```

Result: Logs successfully received in Splunk Search & Reporting

12.4 Data Server Telemetry Sources

Log Source	SIEM	SOC Value
/var/log/auth.log	Wazuh + Splunk	SSH, sudo, AD logins via Kerberos
/var/log/syslog	Wazuh + Splunk	General system activity
/var/log/kern.log	Wazuh + Splunk	Linux kernel events
Wazuh FIM	Wazuh	Server file integrity monitoring
Kerberos tickets	Wazuh	AD user authentication on Linux
PAM events	Wazuh + Splunk	Authorization and access control

12.5 Final Lab Integration Status

ALL SYSTEMS INTEGRATED IN WAZUH AND SPLUNK

System	Wazuh	Splunk	VLAN
USER01 / USER02	ACTIVE	ACTIVE	20 - Users
USER03 / USER04 (EXEC)	ACTIVE	ACTIVE	30 - Executive
DC01 (Windows Server 2025)	ACTIVE	ACTIVE	40 - Servers
Data Server (Ubuntu 24.04)	ACTIVE	ACTIVE	40 - Servers
Kali Blue (jean-blue)	ACTIVE	ACTIVE	50 - SOC
iMac Physical Host (macOS)	ACTIVE	ACTIVE	WAN
Admin Workstation	ACTIVE	ACTIVE	10 - Mgmt

12.6 Portfolio Value

The Data Server integration demonstrates advanced monitoring capabilities in hybrid infrastructure:

- Monitoring of Linux servers joined to Windows Active Directory domain
- Detection of AD authentication on Linux systems via Kerberos
- Visibility of lateral movement from Windows endpoints to Linux servers
- Complete telemetry pipeline across real enterprise infrastructure
- Detection of privilege escalation via sudo on Linux